Yisehak Belay

Mankato, MN 56001
yisehakbelay26@gmail.com | +1 (507) 613-1316 | linkedin.com/in/yisehak-belay | github.com/YisehakBelay

Professional Summary

Computer Information Technology student at Minnesota State University with strong foundations in software development, database systems, and data analytics. Experienced in building distributed systems, REST APIs, full-stack dashboards, and NoSQL pipelines. Delivered a fraud analytics workflow over 13.3M transactions using MongoDB and Power BI. Seeking a Software Engineering or IT Internship to apply technical and problem-solving skills in real-world environments.

Education

Minnesota State University, Mankato — Mankato, MN
B.S. in Computer Information Technology, Math Minor — *Expected Summer 2026*

- **Relevant Coursework:** Python Programming, Data Structures (Java), Algorithms (Python), Computer Architecture, System Analysis & Design, Introduction to Databases, Distributed Database Systems (Oracle), Networking (Cisco), Information Security, Data Analytics (CIS 444)

Technical Skills

Languages & Frameworks: Python, Java, C++, JavaScript (Node.js/Express), TypeScript, SQL, PL/SQL, Next.js
Databases & Data Tools: Oracle DBMS, MongoDB, pymongo, pandas, Power BI
Networking & Infrastructure: Cisco Packet Tracer, VLANs, OSPF, VPN, ASA Firewall
Core Competencies: REST API Design, Distributed Databases, Query Optimization, System Design, Data Pipelines, Aggregation Frameworks, SDLC, UML Modeling

Projects

Fraud Analytics Dashboard | *MongoDB, pymongo, Python, pandas, Power BI, Star Schema* — Spring 2026

- Designed and executed a full data analytics pipeline over a 13.3M-transaction financial dataset (Kaggle star schema: transactions, cards, users) to surface fraud and spending insights for a CIS 444 final project.
- Wrote 10 MongoDB aggregation pipelines (`$lookup`, `$group`, `$match`, `$addFields`) that reduced Power BI payload from 13.3M raw documents to small summarized JSON files, eliminating in-tool performance bottlenecks.
- Built an interactive Power BI dashboard with six visual types (filled map, KPI cards, donut chart, matrix, clustered bar, table), organized into three thematic sections — Customer Segmentation, Geographic Behavior, and Fraud & Risk Signals — answering all 10 defined KPIs.

Distributed Database Management System (DDBMS) | *MySQL, MariaDB Galera Cluster, VLANs, ASA Firewall* — Fall 2025

- Designed and implemented a distributed database system using MySQL and a MariaDB Galera multi-master cluster, achieving ~35% query performance improvement through synchronous replication across 3 global nodes (LA, Tokyo, Buenos Aires).
- Developed data replication and fragmentation strategies that ensured consistency and efficient data access across multiple geographically distributed sites, reducing cross-site query latency.

Student Management REST API | *Node.js, Express, MongoDB, Mongoose* | GitHub — 2025

- Built a production-structured REST API with 28 endpoints across students, courses, enrollments, attendance, and fees modules, enabling full CRUD operations with GPA calculation and payment tracking.
- Modeled a normalized MongoDB schema using Mongoose with relationships across five collections, reducing redundant queries by consolidating enrollment and fee lookups into a single document reference pattern.

Student Dashboard | *TypeScript, Next.js* | GitHub — 2025

- Developed a full-stack student dashboard using TypeScript and Next.js, integrating data from the Student Management REST API to display GPA summaries, enrollment status, and attendance records in a unified UI.

Enterprise Network Simulation | *Cisco Packet Tracer, VLANs, ASA Firewall, OSPF, VPN* — Fall 2025

- Configured enterprise-level network topologies using VLANs, OSPF dynamic routing, and VPN tunnels, improving network segmentation and enabling secure inter-site communication across simulated branch offices.
- Implemented ASA firewall rules and traffic segmentation strategies, reducing the simulated attack surface by restricting inter-VLAN traffic to authorized flows only.

Disney Ride Scheduling Optimization System | *UML, System Design, SDLC* — Spring 2025

- Designed a complete system model for ride scheduling optimization using UML diagrams (use case, class, sequence) and SDLC methodologies, translating business requirements into structured technical specifications.
- Produced stakeholder-ready documentation that bridged business and engineering teams, reducing requirements ambiguity and accelerating design approval cycles.

MongoDB Retail Analytics | *MongoDB, Aggregation Pipeline, Data Modeling* — 2026

- Designed and queried NoSQL databases using MongoDB against real-world retail datasets, applying fact and dimension data modeling concepts to improve query structure and efficiency.
- Built aggregation pipelines to analyze customer behavior, product performance, and sales trends, generating data-driven insights that surface actionable patterns from large-scale retail data.

Experience

Member Dec 2024 – Present
ColorStack *Remote*

- Participate in a national community supporting Black and Latinx computer science students through mentorship programs and collaborative coding workshops, strengthening technical problem-solving and professional networking skills.

Leadership & Involvement

Member 2025 – Present
National Society of Black Engineers (NSBE) *Mankato, MN*
Volunteer Summer 2019
Care Epilepsy Ethiopia